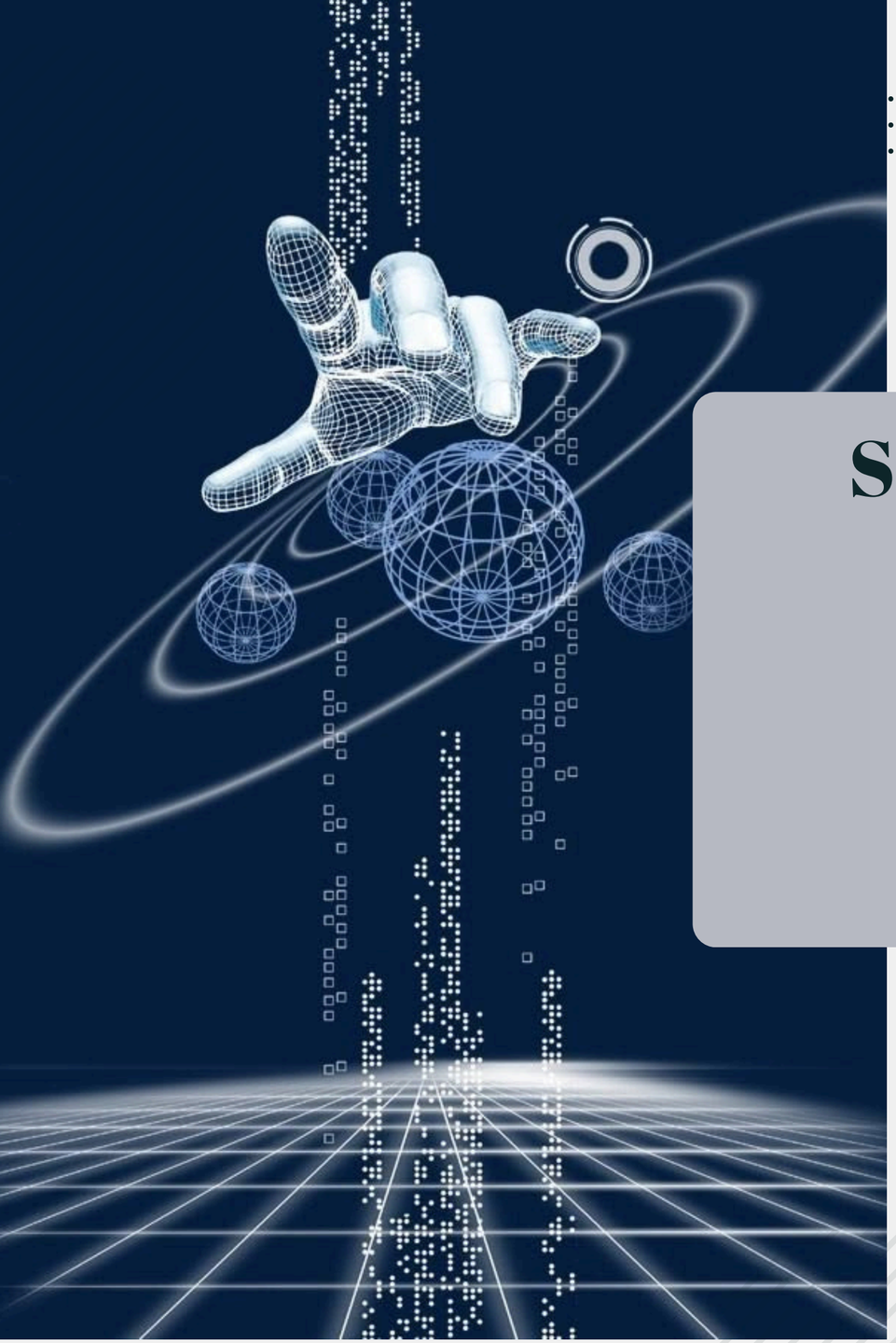


EST. 2024
BRIGHT
WEB CRAFTERS

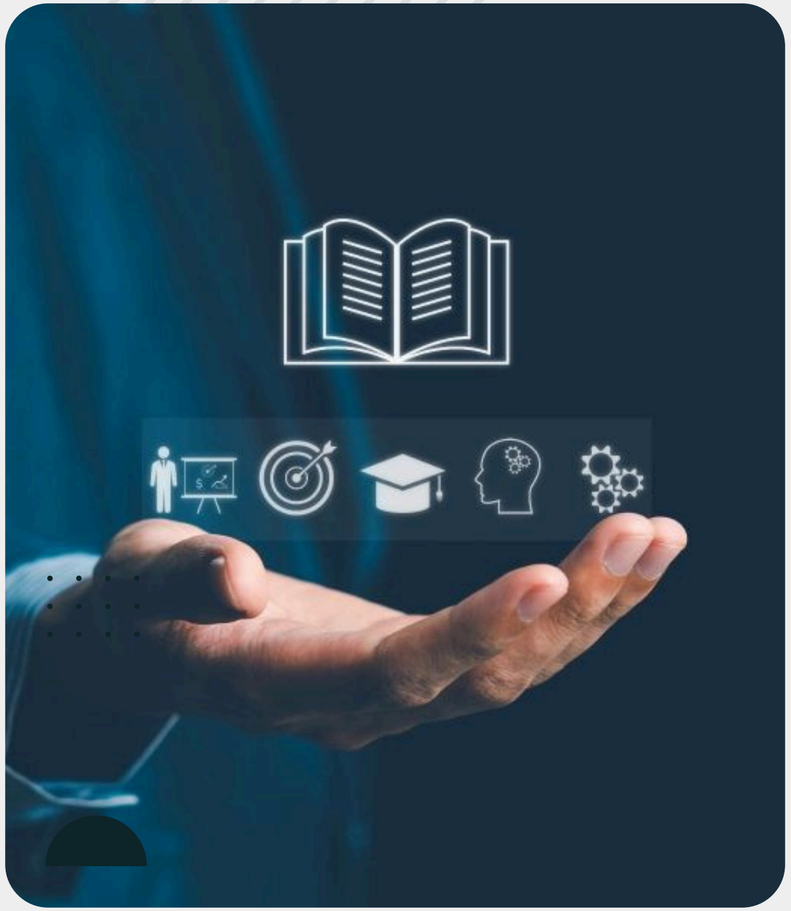
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SCHOOL-PARENTAL SYSTEM

ITC327W

PHASE 3

Introduction

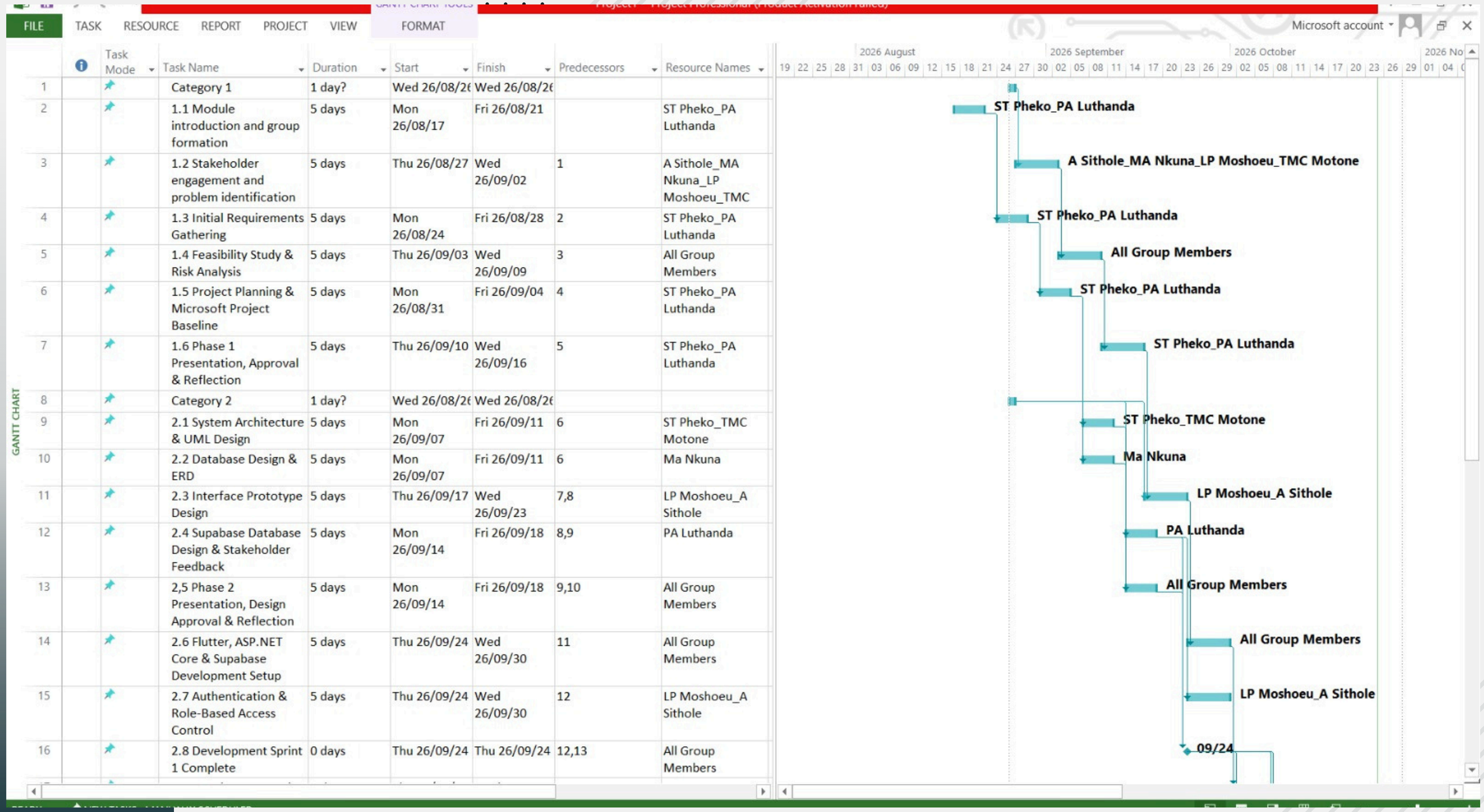


Our project addresses a communication and record-keeping problem we confirmed through interviews at three Welkom schools: Reitzpark Primary, Lenakeng Technical and Thabong Primary.

Schools still rely on paper registers, printed letters and WhatsApp groups. Learners don't always deliver letters, contact numbers are outdated, and schools have no reliable way of knowing whether a parent has received or read an important notice. As a result, parents often find out about attendance or academic problems too late.

Our approved solution is the School-Parental System: a single, secure platform where administrators and teachers use an ASP.NET web application, parents use a Flutter mobile app, and both connect to a shared Supabase backend. The system tracks announcements from unread to read to acknowledged, so the school can confirm that parents have been reached. I'll now demonstrate how it works

MS PROJECT



FILE TASK RESOURCE REPORT PROJECT VIEW FORMAT — Microsoft account

GANTT CHART

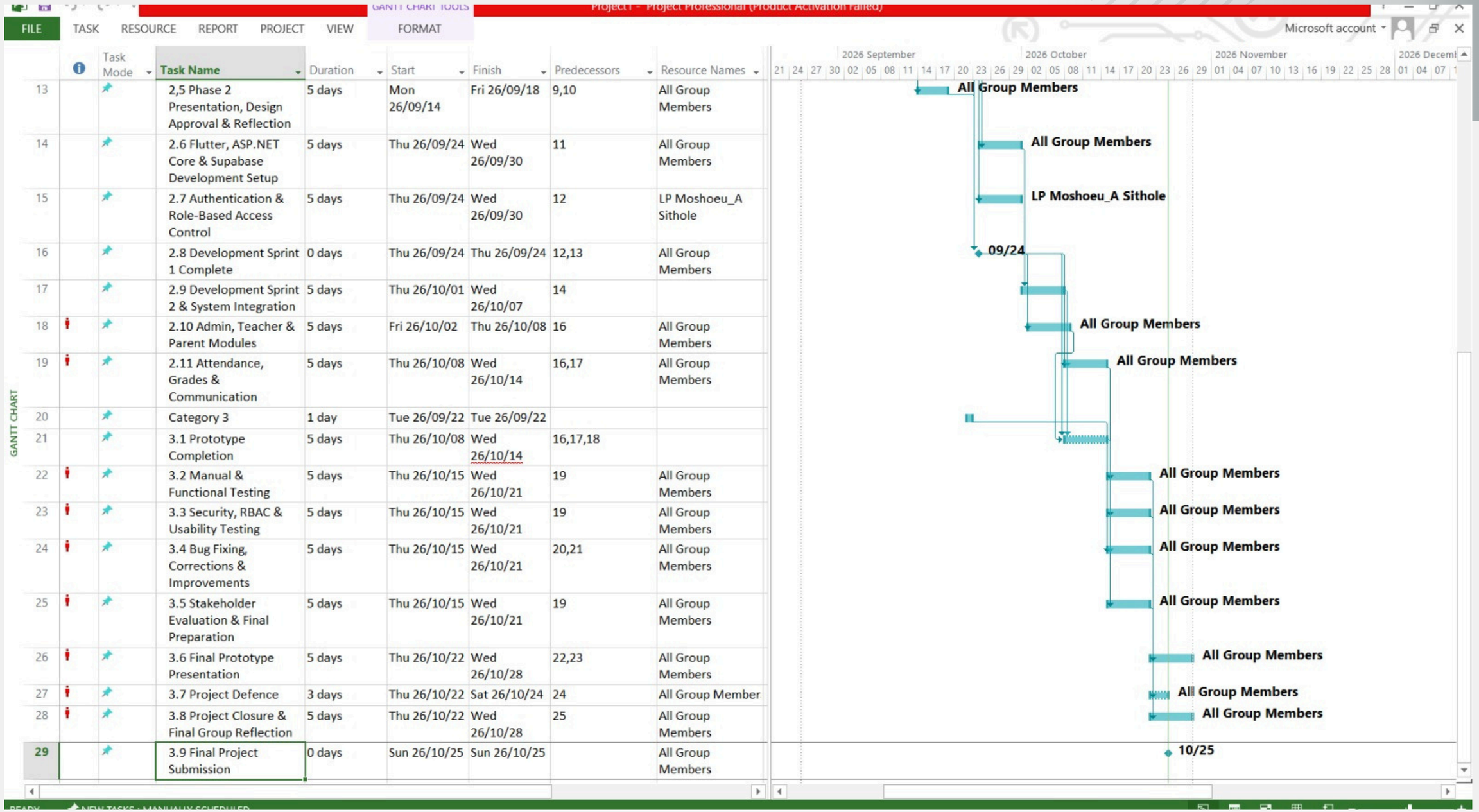
	Task Mode	Task Name	Duration	Start	Finish	Predecessors	Resource Names
1		Category 1	1 day?	Wed 26/08/26	Wed 26/08/26		
2		1.1 Module introduction and group formation	5 days	Mon 26/08/17	Fri 26/08/21		ST Pheko_PA Luthanda
3		1.2 Stakeholder engagement and problem identification	5 days	Thu 26/08/27	Wed 26/09/02	1	A Sithole_MA Nkuna_LP Moshoeu_TMC Motone
4		1.3 Initial Requirements Gathering	5 days	Mon 26/08/24	Fri 26/08/28	2	ST Pheko_PA Luthanda
5		1.4 Feasibility Study & Risk Analysis	5 days	Thu 26/09/03	Wed 26/09/09	3	All Group Members
6		1.5 Project Planning & Microsoft Project Baseline	5 days	Mon 26/08/31	Fri 26/09/04	4	ST Pheko_PA Luthanda
7		1.6 Phase 1 Presentation, Approval & Reflection	5 days	Thu 26/09/10	Wed 26/09/16	5	ST Pheko_PA Luthanda
8		Category 2	1 day?	Wed 26/08/26	Wed 26/08/26		
9		2.1 System Architecture & UML Design	5 days	Mon 26/09/07	Fri 26/09/11	6	ST Pheko_TMC Motone
10		2.2 Database Design & ERD	5 days	Mon 26/09/07	Fri 26/09/11	6	Ma Nkuna
11		2.3 Interface Prototype Design	5 days	Thu 26/09/17	Wed 26/09/23	7,8	LP Moshoeu_A Sithole
12		2.4 Supabase Database Design & Stakeholder Feedback	5 days	Mon 26/09/14	Fri 26/09/18	8,9	PA Luthanda
13		2,5 Phase 2 Presentation, Design Approval & Reflection	5 days	Mon 26/09/14	Fri 26/09/18	9,10	All Group Members
14		2.6 Flutter, ASP.NET Core & Supabase Development Setup	5 days	Thu 26/09/24	Wed 26/09/30	11	All Group Members
15		2.7 Authentication & Role-Based Access Control	5 days	Thu 26/09/24	Wed 26/09/30	12	LP Moshoeu_A Sithole
16		2.8 Development Sprint 1 Complete	0 days	Thu 26/09/24	Thu 26/09/24	12,13	All Group Members

2026 August | 2026 September | 2026 October | 2026 No

19 22 25 28 31 03 06 09 12 15 18 21 24 27 30 02 05 08 11 14 17 20 23 26 29 02 05 08 11 14 17 20 23 26 29 01 04

ST Pheko_PA Luthanda
A Sithole_MA Nkuna_LP Moshoeu_TMC Motone
ST Pheko_PA Luthanda
All Group Members
ST Pheko_PA Luthanda
ST Pheko_PA Luthanda
ST Pheko_TMC Motone
Ma Nkuna
LP Moshoeu_A Sithole
PA Luthanda
All Group Members
All Group Members
LP Moshoeu_A Sithole
09/24

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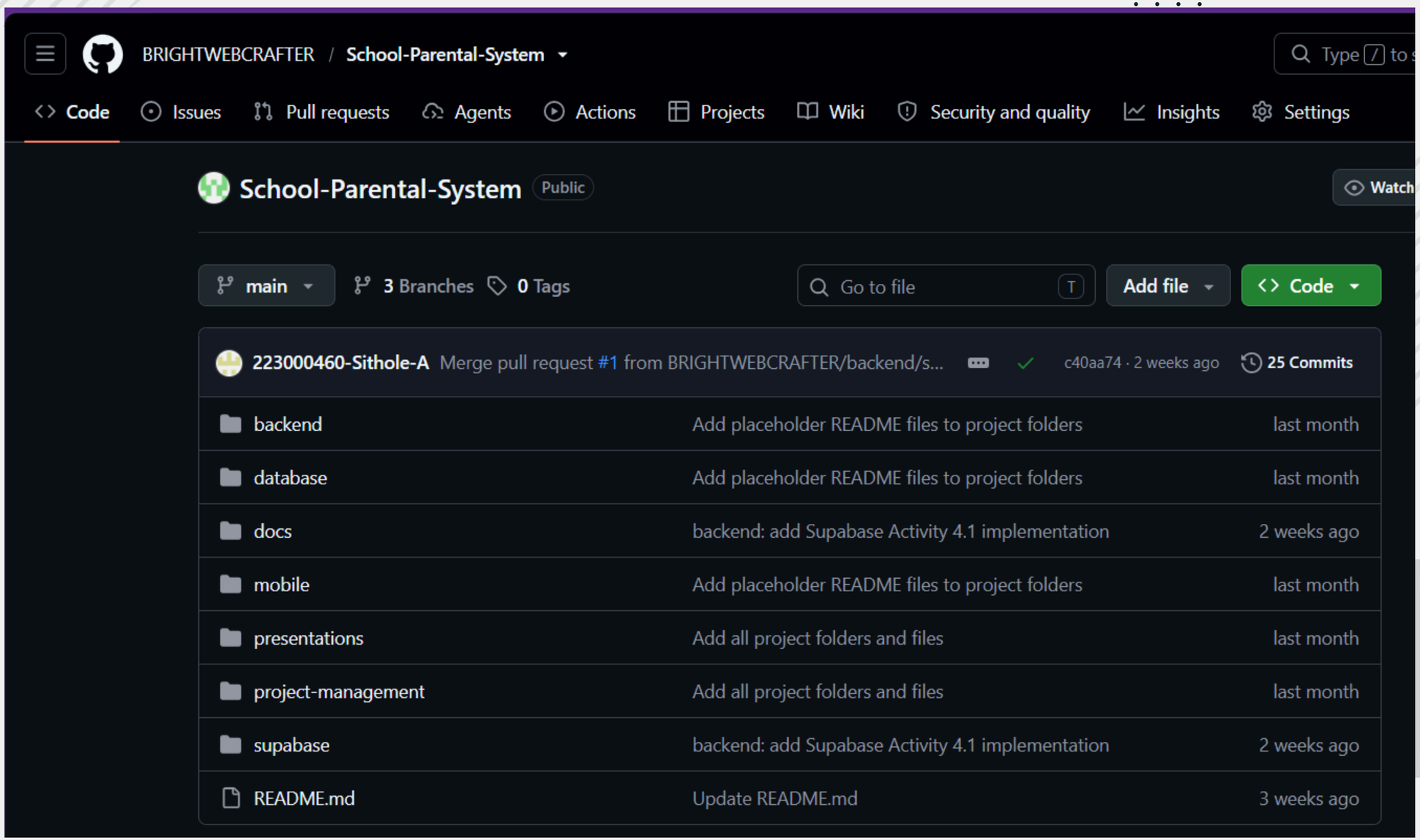


	Task Mode	Task Name	Duration	Start	Finish	Predecessors	Resource Names
13		2,5 Phase 2 Presentation, Design Approval & Reflection	5 days	Mon 26/09/14	Fri 26/09/18	9,10	All Group Members
14		2.6 Flutter, ASP.NET Core & Supabase Development Setup	5 days	Thu 26/09/24	Wed 26/09/30	11	All Group Members
15		2.7 Authentication & Role-Based Access Control	5 days	Thu 26/09/24	Wed 26/09/30	12	LP Moshoeu_A Sithole
16		2.8 Development Sprint 1 Complete	0 days	Thu 26/09/24	Thu 26/09/24	12,13	All Group Members
17		2.9 Development Sprint 2 & System Integration	5 days	Thu 26/10/01	Wed 26/10/07	14	
18		2.10 Admin, Teacher & Parent Modules	5 days	Fri 26/10/02	Thu 26/10/08	16	All Group Members
19		2.11 Attendance, Grades & Communication	5 days	Thu 26/10/08	Wed 26/10/14	16,17	All Group Members
20		Category 3	1 day	Tue 26/09/22	Tue 26/09/22		
21		3.1 Prototype Completion	5 days	Thu 26/10/08	Wed 26/10/14	16,17,18	
22		3.2 Manual & Functional Testing	5 days	Thu 26/10/15	Wed 26/10/21	19	All Group Members
23		3.3 Security, RBAC & Usability Testing	5 days	Thu 26/10/15	Wed 26/10/21	19	All Group Members
24		3.4 Bug Fixing, Corrections & Improvements	5 days	Thu 26/10/15	Wed 26/10/21	20,21	All Group Members
25		3.5 Stakeholder Evaluation & Final Preparation	5 days	Thu 26/10/15	Wed 26/10/21	19	All Group Members
26		3.6 Final Prototype Presentation	5 days	Thu 26/10/22	Wed 26/10/28	22,23	All Group Members
27		3.7 Project Defence	3 days	Thu 26/10/22	Sat 26/10/24	24	All Group Member
28		3.8 Project Closure & Final Group Reflection	5 days	Thu 26/10/22	Wed 26/10/28	25	All Group Members
29		3.9 Final Project Submission	0 days	Sun 26/10/25	Sun 26/10/25		All Group Members

progress tracker

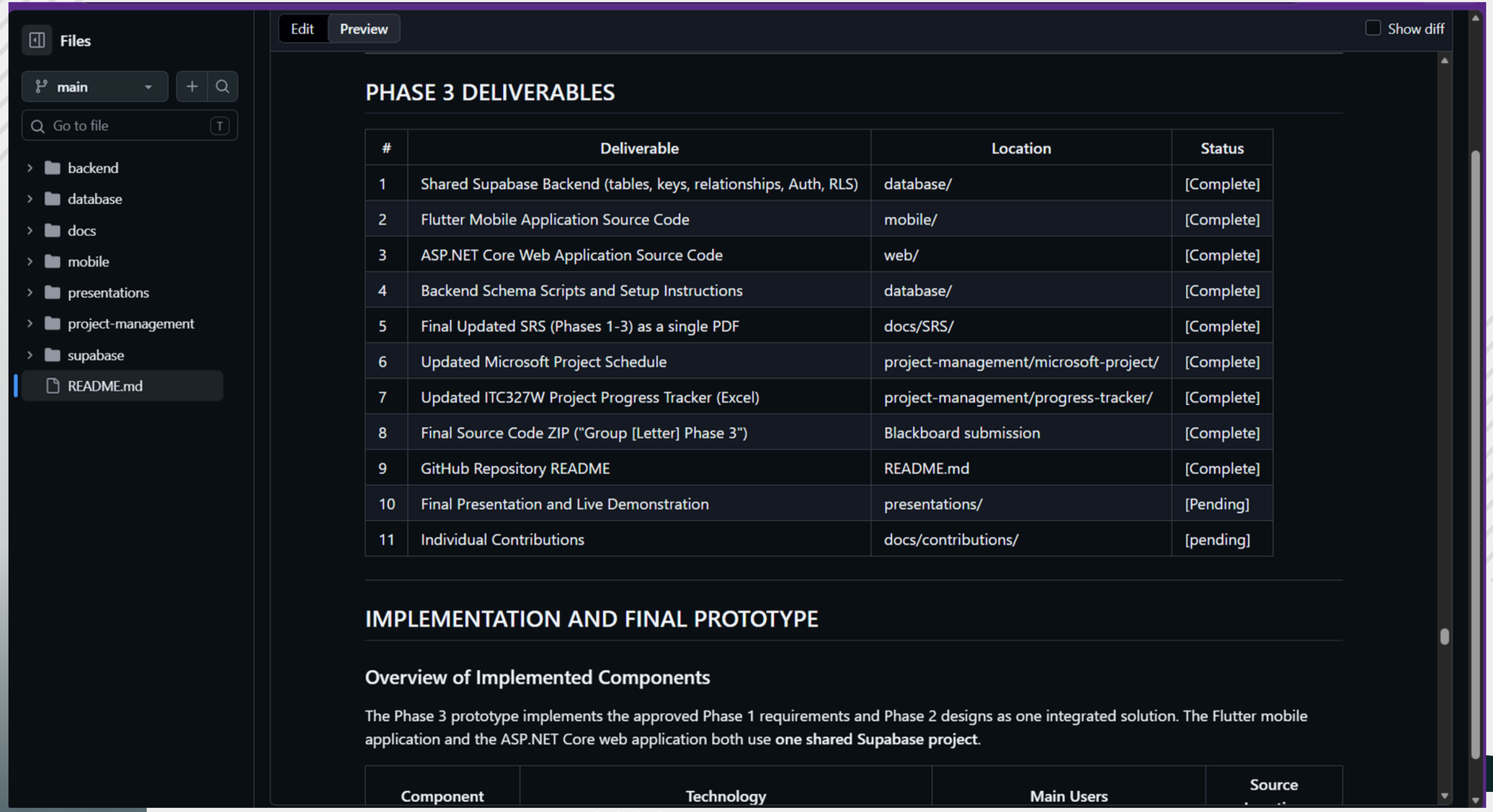
	A	B	C	D	E	F	G	H	I	J	K	L
1	**Individual Contribution Record**											
2	**Student Name**	**Student Number**	**Reporting Phase**	**Activity**	**Assigned Responsibility**	**Work Personally Completed**	**Date Completed**	**Hours Spent**	**Evidence / Link**	**Verified by Group**	**Verification Date**	**Notes**
3	LP Moshoeu	223046876	Phase 2	Activity 3.3	Figma(Mobile Application)	Do The figma mobile application	25/09/2026	130,0	https://www.figma.com/proto/pG7mL3	Yes	25/09/2026	
4	A Sithole	223000460	Phase 2	Activity 3.3	Figma(Web)	To do the figma web	25/09/2026	132,0	https://www.figma.com/proto/IPObV3	Yes	25/09/2026	
5	MA Nkuna	224000274	Phase 2	Activity 3.2	ERD	To make our ERD	25/09/2026	62,0		Yes	25/09/2026	
6	ST Pheko	223050336	Phase 2	Activity 3.1	System Architecture	To make the system Archicture	25/09/2026	60,0	https://viewer.diagrams.net/?tags=%7	Yes	25/09/2026	
7	TMC Motone	224027806	Phase 2	Activity 3.2	Sequence Diagram	To draw up our Sequence Diagram	25/09/2026	55,0		Yes	25/09/2026	
8	PA Luthanda	222023335	Phase 2	Activity 3.2	Use Case Diagram	To make our use case	25/09/2026	53,0	https://viewer.diagrams.net/?tags=%7	Yes	25/09/2026	
9												
10	LP Moshoeu	223046876	Phase 3	Sections 1 & 2	Flutter and supabase	Developing some functions of flutter and	25/10/2026	90,0	Submitted our project	Yes		
11	A Sithole	223000460	Phase 3	Sections 1 & 3	ASP.NET and supabase	Developing some functions of our ASP.NET	25/10/2026	150,0	Submitted our project	Yes		
12	MA Nkuna	224000274	Phase 3	Sections 1 & 3	ASP.NET and supabase	Developing some functions of our ASP.NET	25/10/2026	150,0	Submitted our project	Yes		
13	ST Pheko	223050336	Phase 3	Sections 1 & 2	Flutter and supabase	Developing some functions of flutter and	25/10/2026	83,0	Submitted our project	Yes		
14	TMC Motone	224027806	Phase 3	Sections 1 & 3	ASP.NET and supabase	Developing some functions of our ASP.NET	25/10/2026	79,0	Submitted our project	Yes		
15	PA Luthanda	222023335	Phase 3	Sections 1 & 2	Flutter and supabase	Developing some functions of flutter and	25/10/2026	81,0	Submitted our project	Yes		
16												
17												
18												
19												

GITHUB


BRIGHTWEBCRAFTER / School-Parental-System
Type / to
Code
Issues
Pull requests
Agents
Actions
Projects
Wiki
Security and quality
Insights
Settings
School-Parental-System
Public
Watch
main
3 Branches
0 Tags
Go to file
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Add file
Code
223000460-Sithole-A Merge pull request #1 from BRIGHTWEBCRAFTER/backend/s...
c40aa74 · 2 weeks ago
25 Commits
backend
Add placeholder README files to project folders
last month
database
Add placeholder README files to project folders
last month
docs
backend: add Supabase Activity 4.1 implementation
2 weeks ago
mobile
Add placeholder README files to project folders
last month
presentations
Add all project folders and files
last month
project-management
Add all project folders and files
last month
supabase
backend: add Supabase Activity 4.1 implementation
2 weeks ago
README.md
Update README.md
3 weeks ago

CONT.


Files
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database
docs
mobile
presentations
project-management
supabase
README.md
Edit
Preview
Show diff
PHASE 3 - SYSTEM DEVELOPMENT, IMPLEMENTATION AND FINAL DEMONSTRATION
PROJECT PROGRESS - PHASE 3
Completed
Phase 1 SRS (requirements, stakeholder engagement, feasibility)
Phase 2 System Design (architecture, UML, ERD, wireframes, design alignment)
Supabase backend implementation (tables, relationships, Authentication)
Flutter mobile application development
ASP.NET Core web application development
Integration of Flutter and ASP.NET Core with one shared Supabase project
Final Microsoft Project schedule update
Final SRS update (Implementation and Final Prototype section)
Remaining / Outstanding
[LIST ANY INCOMPLETE FUNCTIONS HERE, or write "None"]
Final presentation and live demonstration
Individual contribution check (submitted separately by each student)
PHASE 3 DELIVERABLES
#
Deliverable
Location
Status



PHASE 3 DELIVERABLES

#	Deliverable	Location	Status
1	Shared Supabase Backend (tables, keys, relationships, Auth, RLS)	database/	[Complete]
2	Flutter Mobile Application Source Code	mobile/	[Complete]
3	ASP.NET Core Web Application Source Code	web/	[Complete]
4	Backend Schema Scripts and Setup Instructions	database/	[Complete]
5	Final Updated SRS (Phases 1-3) as a single PDF	docs/SRS/	[Complete]
6	Updated Microsoft Project Schedule	project-management/microsoft-project/	[Complete]
7	Updated ITC327W Project Progress Tracker (Excel)	project-management/progress-tracker/	[Complete]
8	Final Source Code ZIP ("Group [Letter] Phase 3")	Blackboard submission	[Complete]
9	GitHub Repository README	README.md	[Complete]
10	Final Presentation and Live Demonstration	presentations/	[Pending]
11	Individual Contributions	docs/contributions/	[pending]

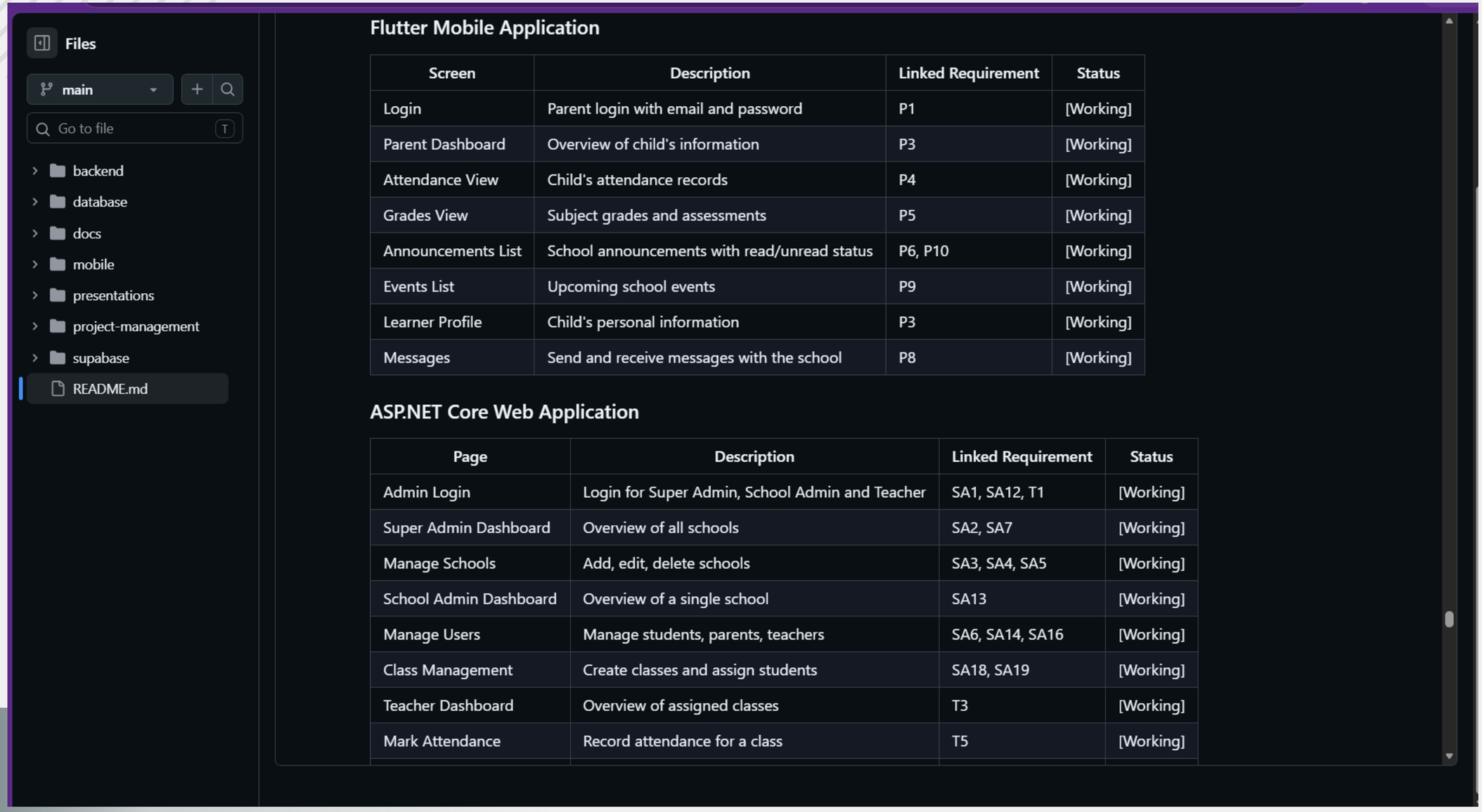
IMPLEMENTATION AND FINAL PROTOTYPE

Overview of Implemented Components

The Phase 3 prototype implements the approved Phase 1 requirements and Phase 2 designs as one integrated solution. The Flutter mobile application and the ASP.NET Core web application both use **one shared Supabase project**.

Component	Technology	Main Users	Source

CONT.



Flutter Mobile Application

Screen	Description	Linked Requirement	Status
Login	Parent login with email and password	P1	[Working]
Parent Dashboard	Overview of child's information	P3	[Working]
Attendance View	Child's attendance records	P4	[Working]
Grades View	Subject grades and assessments	P5	[Working]
Announcements List	School announcements with read/unread status	P6, P10	[Working]
Events List	Upcoming school events	P9	[Working]
Learner Profile	Child's personal information	P3	[Working]
Messages	Send and receive messages with the school	P8	[Working]

ASP.NET Core Web Application

Page	Description	Linked Requirement	Status
Admin Login	Login for Super Admin, School Admin and Teacher	SA1, SA12, T1	[Working]
Super Admin Dashboard	Overview of all schools	SA2, SA7	[Working]
Manage Schools	Add, edit, delete schools	SA3, SA4, SA5	[Working]
School Admin Dashboard	Overview of a single school	SA13	[Working]
Manage Users	Manage students, parents, teachers	SA6, SA14, SA16	[Working]
Class Management	Create classes and assign students	SA18, SA19	[Working]
Teacher Dashboard	Overview of assigned classes	T3	[Working]
Mark Attendance	Record attendance for a class	T5	[Working]

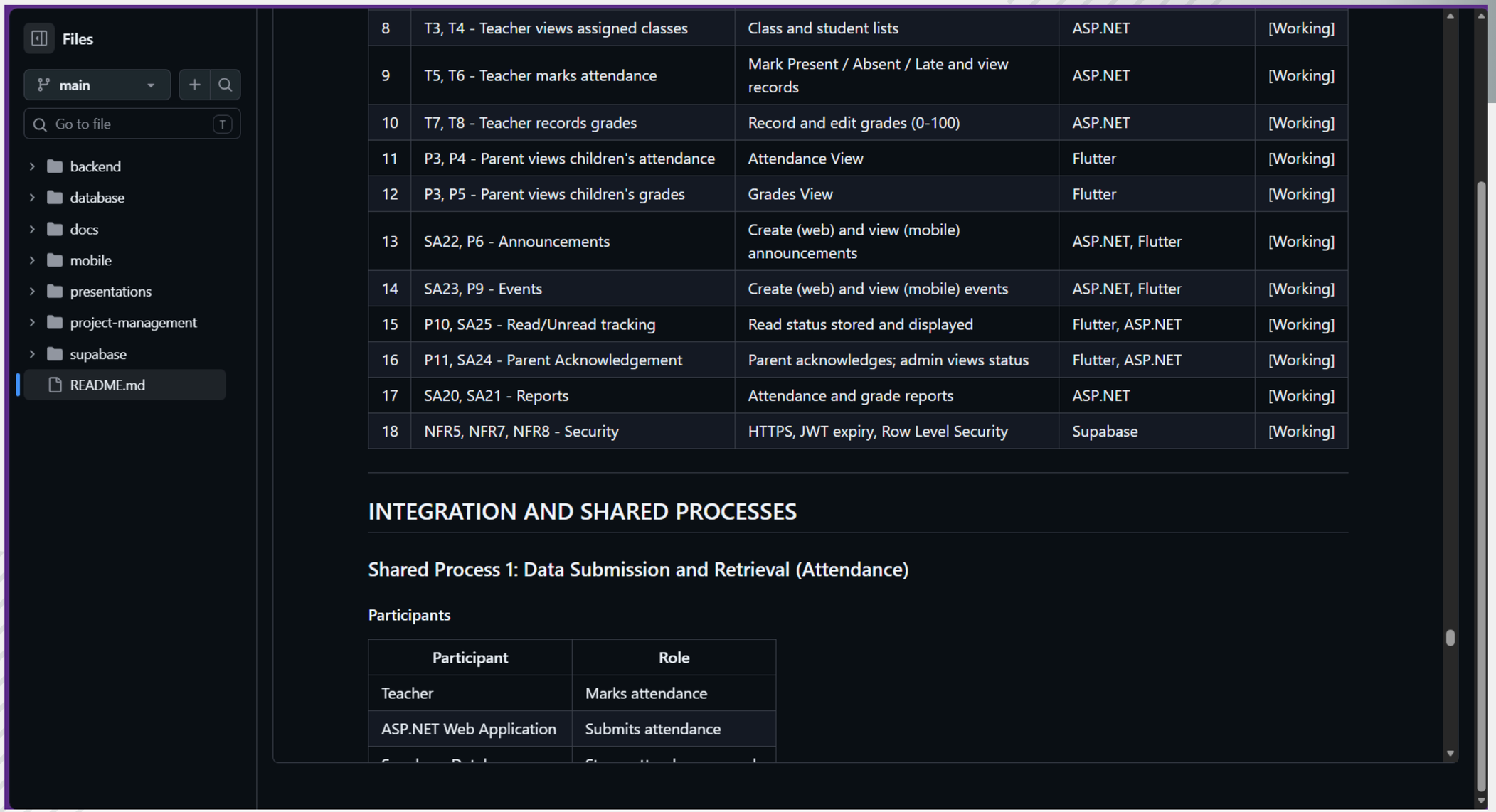


Teacher Dashboard	Overview of assigned classes	T3	[Working]
Mark Attendance	Record attendance for a class	T5	[Working]
Record Grades	Record and edit grades for students	T7, T8	[Working]
Create Announcement	Create and target announcements	SA22	[Working]
Create Event	Create school events	SA23	[Working]
Reports	Attendance and grade reports	SA20, SA21	[Working]
Messages	View and respond to messages	SA25	[Working]

REQUIREMENTS TO IMPLEMENTATION TABLE

#	Approved Requirement	Implemented Function	Application	Status
1	SA1, SA12, T1, P1 - Users can register and login	Email/password login using Supabase Auth	Flutter, ASP.NET	[Working]
2	Role-Based Access Control	Role checks and Row Level Security policies	Flutter, ASP.NET, Supabase	[Working]
3	SA3-SA5 - Super Admin manages schools	Add, edit and delete schools	ASP.NET	[Working]
4	SA8 - Super Admin selects a school	School selector (Super Admin only)	ASP.NET	[Working]
5	SA14, SA15 - School Admin manages students	Add and edit students	ASP.NET	[Working]
6	SA16, SA17 - School Admin manages teachers	Add and edit teachers	ASP.NET	[Working]
7	SA18, SA19 - Classes	Create classes and assign students	ASP.NET	[Working]
8	T3, T4 - Teacher views assigned classes	Class and student lists	ASP.NET	[Working]

CONT.



8	T3, T4 - Teacher views assigned classes	Class and student lists	ASP.NET	[Working]
9	T5, T6 - Teacher marks attendance	Mark Present / Absent / Late and view records	ASP.NET	[Working]
10	T7, T8 - Teacher records grades	Record and edit grades (0-100)	ASP.NET	[Working]
11	P3, P4 - Parent views children's attendance	Attendance View	Flutter	[Working]
12	P3, P5 - Parent views children's grades	Grades View	Flutter	[Working]
13	SA22, P6 - Announcements	Create (web) and view (mobile) announcements	ASP.NET, Flutter	[Working]
14	SA23, P9 - Events	Create (web) and view (mobile) events	ASP.NET, Flutter	[Working]
15	P10, SA25 - Read/Unread tracking	Read status stored and displayed	Flutter, ASP.NET	[Working]
16	P11, SA24 - Parent Acknowledgement	Parent acknowledges; admin views status	Flutter, ASP.NET	[Working]
17	SA20, SA21 - Reports	Attendance and grade reports	ASP.NET	[Working]
18	NFR5, NFR7, NFR8 - Security	HTTPS, JWT expiry, Row Level Security	Supabase	[Working]

INTEGRATION AND SHARED PROCESSES

Shared Process 1: Data Submission and Retrieval (Attendance)

Participants

Participant	Role
Teacher	Marks attendance
ASP.NET Web Application	Submits attendance

Shared Process 1: Data Submission and Retrieval (Attendance)

Participants

Participant	Role
Teacher	Marks attendance
ASP.NET Web Application	Submits attendance
Supabase Database	Stores attendance records
Flutter Mobile Application	Retrieves attendance
Parent	Views attendance

Workflow

1. Teacher logs into the ASP.NET web application
2. Supabase Auth validates the credentials and returns a JWT token
3. Teacher selects an assigned class and marks each student as Present, Absent, or Late
4. Teacher clicks Save Attendance
5. The application validates the teacher's role and class assignment
6. Attendance records are saved to the Attendance table in Supabase
7. Parent logs into the Flutter mobile application
8. Parent opens Attendance View
9. Flutter retrieves the attendance records from Supabase (Row Level Security limits results to the parent's own children)
10. The same attendance values saved by the teacher are displayed to the parent

Requirements Alignment

Requirement	Use Case

Files

main

Go to file T

- backend
- database
- docs
- mobile
- presentations
- project-management
- supabase
- README.md

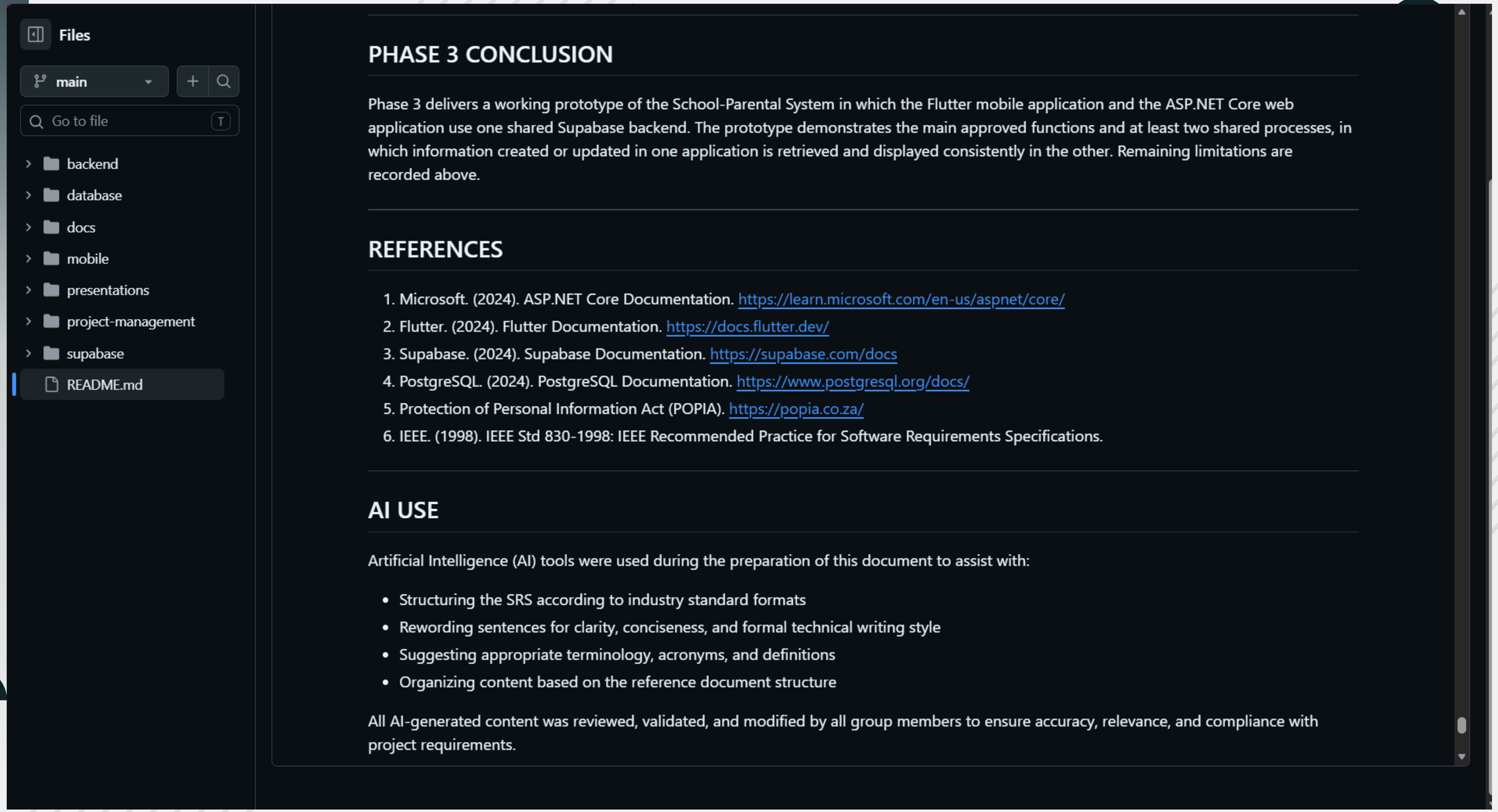
MICROSOFT PROJECT PLAN - PHASE 3

Project Schedule Screenshot

[INSERT FINAL MICROSOFT PROJECT SCREENSHOT HERE]

Phase 3 Activities

Task	Activity	Start Date	End Date	Resource	Status
2.6	Flutter, ASP.NET Core and Supabase Development Setup	24 Sep 2026	30 Sep 2026	All Group Members	[Complete]
2.7	Authentication and Role-Based Access Control	24 Sep 2026	30 Sep 2026	LP Moshoeu, A Sithole	[Complete]
2.8	Development Sprint 1 Complete (milestone)	24 Sep 2026	24 Sep 2026	All Group Members	[Complete]
2.9	Development Sprint 2 and System Integration	1 Oct 2026	7 Oct 2026		[Complete]
2.10	Admin, Teacher and Parent Modules	2 Oct 2026	8 Oct 2026	All Group Members	[Complete]
2.11	Attendance, Grades and Communication	8 Oct 2026	14 Oct 2026	All Group Members	[Complete]
3.1	Prototype Completion	8 Oct 2026	14 Oct 2026		[Complete]
3.2	Manual and Functional Testing	15 Oct 2026	21 Oct 2026	All Group Members	[Complete]
3.3	Security, RBAC and Usability Testing	15 Oct 2026	21 Oct 2026	All Group Members	[Complete]
3.4	Bug Fixing, Corrections and Improvements	15 Oct 2026	21 Oct 2026	All Group Members	[Complete]
3.5	Stakeholder Evaluation and Final Preparation	15 Oct 2026	21 Oct 2026	All Group Members	[Complete]
3.6	Final Prototype Presentation	22 Oct 2026	28 Oct 2026	All Group Members	[Pending]
3.7	Project Defence	22 Oct 2026	24 Oct 2026	All Group Members	[Pending]
3.8	Project Closure and Final Group Reflection	22 Oct 2026	28 Oct 2026	All Group Members	[Pending]



PHASE 3 CONCLUSION

Phase 3 delivers a working prototype of the School-Parental System in which the Flutter mobile application and the ASP.NET Core web application use one shared Supabase backend. The prototype demonstrates the main approved functions and at least two shared processes, in which information created or updated in one application is retrieved and displayed consistently in the other. Remaining limitations are recorded above.

REFERENCES

1. Microsoft. (2024). ASP.NET Core Documentation. https://learn.microsoft.com/en-us/aspnet/core/
2. Flutter. (2024). Flutter Documentation. https://docs.flutter.dev/
3. Supabase. (2024). Supabase Documentation. https://supabase.com/docs
4. PostgreSQL. (2024). PostgreSQL Documentation. https://www.postgresql.org/docs/
5. Protection of Personal Information Act (POPIA). https://popia.co.za/
6. IEEE. (1998). IEEE Std 830-1998: IEEE Recommended Practice for Software Requirements Specifications.

AI USE

Artificial Intelligence (AI) tools were used during the preparation of this document to assist with:

- Structuring the SRS according to industry standard formats
- Rewording sentences for clarity, conciseness, and formal technical writing style
- Suggesting appropriate terminology, acronyms, and definitions
- Organizing content based on the reference document structure

All AI-generated content was reviewed, validated, and modified by all group members to ensure accuracy, relevance, and compliance with project requirements.

QUESTIONS

